Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-
"To eliminate concern of Mirror patch screw slip from Scorpio all models"

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio –Xylo TCF.

	Plan	Actual
Start Date	**01.06.19**	**30.06.19**
Completion Date	**20.01.20**	**20.01.20**

Current Status: **CLOSED**

Cross Functional Team

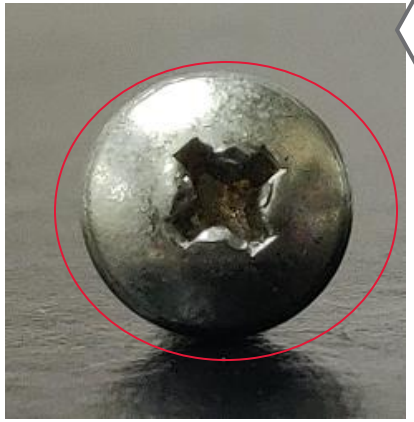
No	Description	Name	Designation	Dept	Phone
1	**Team Leader**	Santosh Jadhav	L-7 0	STCF	9423309400
2	**Mentor**	Rahul Narkar	L-5 DH	STCF	9881080299
3	**Member 1**	Chikhale Sandip	L-5 M	STCF	9922408138
4	**Member 2**	Vivek Kumar	L-8 O	PVT	8825370539
5	**Member 3**	Sanket Pimplapure	L-7 O	SQA	9503500998
6	**Member 4**	Ganesh Bendale	L-7 O	STCF	9850952219
7	**Member 5**	Thakkar Darshak	L-7 O	SQA	7720074994

Step1. Select Subject

Brief Description :

- **What :** Mirror patch screw getting slip.
- **Where:** At mirror patch assembly stage..
- **When:** Since Jun 2019
- **Who:** by stage associate.
- **Why:** Repeat FRC loss / FBO loss concern ,Rework is critical, Screw bit frequent replacement.
- **How:** Average 40 DPT / month , Daily 9-10 cases.

OK condition of Screw head



NOT OK condition of screw head after slip

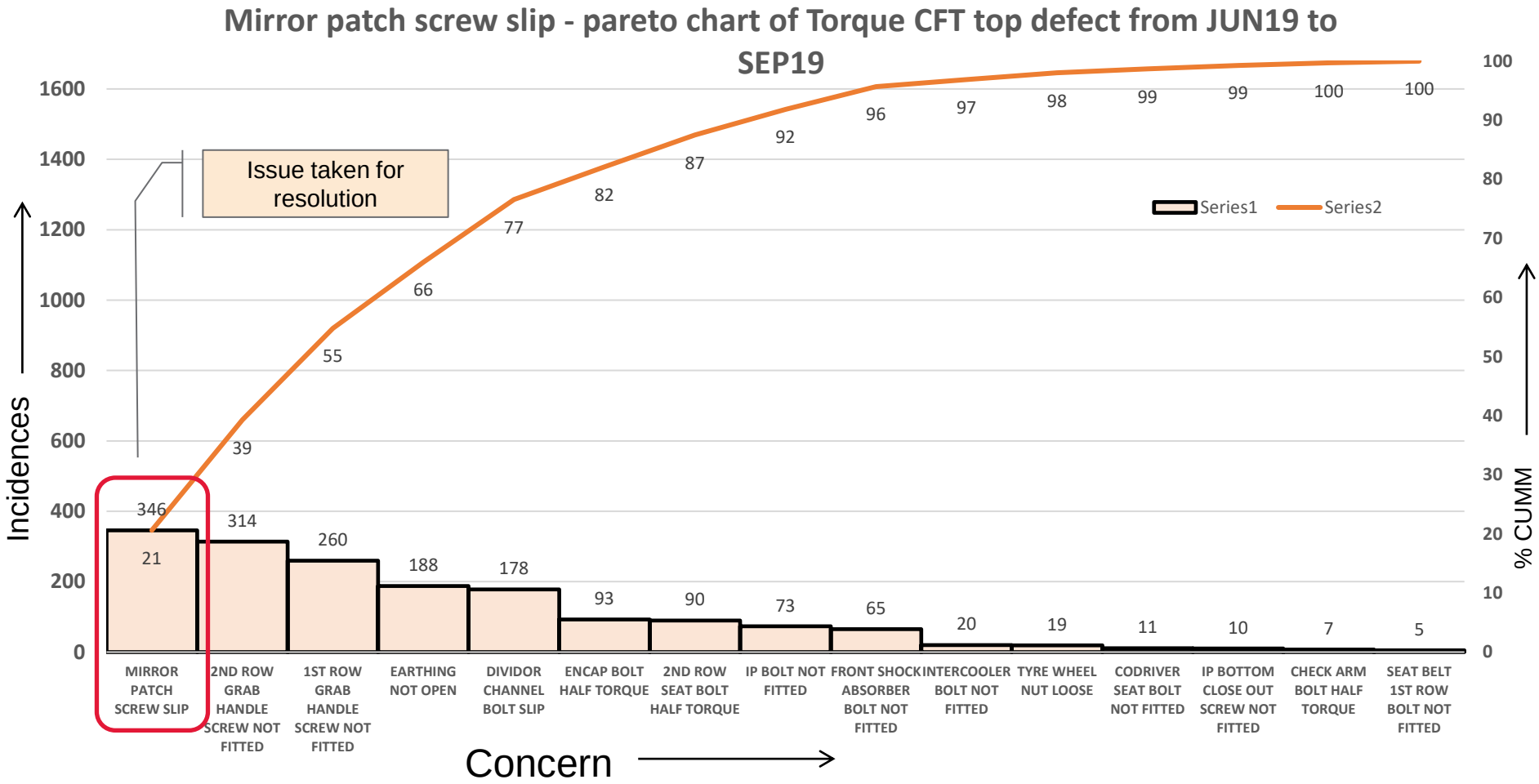
2. Justify the choice (reasons)

Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

2. Justify the choice

Pareto chart :-



INFERENCE :-

Mirror patch screw slip is top concern in TCF in Torque CFT from JUN19 to Sep 19.

2. Justify the choice

Cost of Poor Quality

Tightening gun bit :35,640 Rs/-

Cost of 1 Screw bit = 99 Rs per piece
Bit consumption increased by 30 nos per month
Total cost = 99 *30*12 = 35,640 Rs /-
Rejection projection in one year =35640 Rs / Year

Rework Cost estimation for handbrake bolt half torque-		
Description	Units	
Rework time (Approx)	Min	10
Avg Concerns/Day	Nos	10
Total Required Rework time per day	Min	100
Effective working time	Min	465
Operator required /day	nos	0.22
Approx Operator cost/ day	Rs	1500
Operator cost /concern	Rs	322.58
Avg no. of working days	days	25
Operator cost /month	Rs	8064.52
Operator Cost/year	**Rs**	**96774.19**
Screw bit cost	Rs / month.	2970
yearly cost Tap cost	Rs / month.	35640
Total Rework cost	Rs / Year	132414

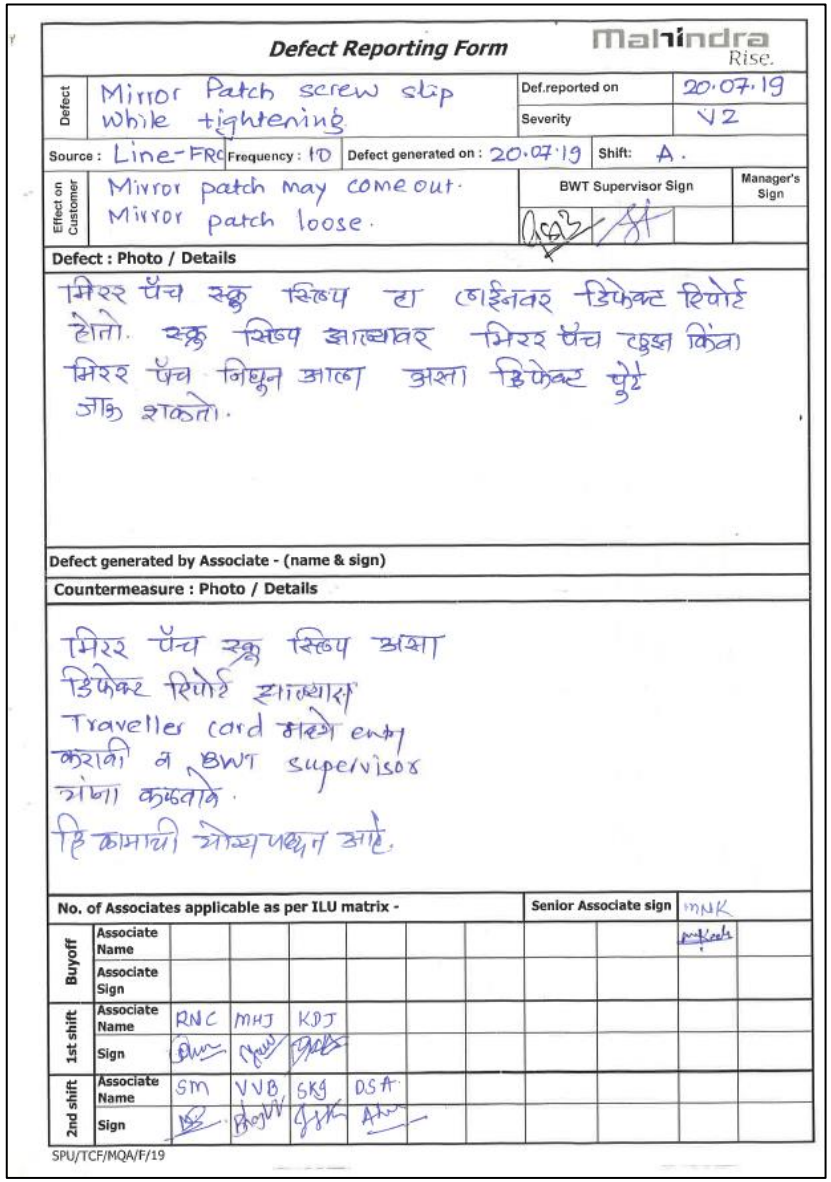
Total Cost/ year = Material cost + Manpower cost = 35640+96114 = 1,32,414/-

2.b Immediate Containment Action

Immediate containment action:-

1. DRF to stage operator and buy off check man to detect defect.

3.Awareness given to stage associates for rework of screw slip issue.



Defect Reporting Form — Mahindra Rise.

Defect	Mirror Patch screw slip while tightening	Def.reported on	20.07.19
		Severity	V2
Source : Line-FRC	Frequency : 10	Defect generated on : 20.07.19	Shift: A.
Effect on Customer	Mirror patch may come out. Mirror patch loose.	BWT Supervisor Sign	Manager's Sign

Defect : Photo / Details

मिरर पॅच स्क्रू स्लिप हा कोईनवर डिफेक्ट रिपोर्ट होतो. स्क्रू स्लिप झाल्यावर मिरर पॅच ढिला किंवा मिरर पॅच निघून आला असा डिफेक्ट पुढे जाऊ शकतो.

Defect generated by Associate - (name & sign)

Countermeasure : Photo / Details

मिरर पॅच स्क्रू स्लिप असा डिफेक्ट रिपोर्ट झाल्यास Traveller card मध्ये entry करावी व BWT supervisor यांना कळवावे.
हि कामाची योग्य पध्दत आहे.

No. of Associates applicable as per ILU matrix - | **Senior Associate sign** MNK

Buyoff	Associate Name										
	Associate Sign										
1st shift	Associate Name	RNC	MHJ	KDJ							
	Sign										
2nd shift	Associate Name	SM	VVB	SKJ	DSA						
	Sign										

SPU/TCF/MQA/F/19

Step 3. Understanding Current Situation

Mahindra

Conformity Analysis
(L0 Analysis)

PU/ Area : Scorpio / Xylo TCF
Cell / Line : Line
BWT : 5

Requested By : MQA	**Date - 01.08.2019**	**Source - Line**	Conf. Analysis Sr.No :

Problem / Concern Statement : Mirror patch screw slip at Trim3

FMEA Severity - **Gravity** : ☐ V1+ ☑ V1 ☐ V2 ☐ V3	**No of Incidences on the day (I) : 30Nos.**	**Vehicle/ Aggre. No. & model - Scorpio W110**	**Concern Priority Number :**

Date of Incidence : 01.08.209	Shift / Time :1st / 3 pm	**Whether SC/CC/Z parameter affected ?**	☐ YES ☐ NO ☐ Z ☐ CC ☐ SC

Immediate containment action for Customer Protection :
WIP,CIA area checked. Rework started screw replaced with new screw

Does a related Conformity Analysis or QC story already exist ?	☐ YES ☑ NO	If Yes	Conformity Analysis No QC Story Sr. No

Failure mode or cause/s identified - 1 . Problem follows with part.

Problem Assigned to - SANTOSH JADHAV	BWT : 04

Stage / Post Analysis -

Does a control already exist ? ☑ YES ☐ NO

Man (Human actions)

Operator name - N D NIKAM
Category - ☑ DR ☐ N1 ☐ EPP ☐ APP
Is operator trained on the stage ? ☑ YES ☐ NO
Operator level (I.L.U) & Dexterity level(1..4) | I | 3 |
Is operator under training ? ☐ under trg.on stage
☐ QC Stage ☐ Redeployed after 1 month ☑ 3*3 matrix completed
Bad rework ☐ YES ☑ NO Newly deployed for rework - Yes **Buyoff opr.name - N / A**

Is the specified facility used ? ☑ YES ☐ NO
Last calibration dates
Facility in good condition ? ☑ YES ☐ NO
Is it a substitute facility ? ☑ YES ☐ NO
Are machine parameters known ? ☐ YES ☑ NO
Parameters are respected ? ☑ YES ☑ NO

Machine

Method (Process)

Main steps are respected ? ☑ YES ☐ NO
Is there a related key point in SOP/CP/SOS ? ☐ YES ☑ NO
Are the reasons for key point known ? ☑ YES ☐ NO
Are operating reqmnts respected ? ☑ YES ☐ NO
Is the defect part of the checkman/ Buy off list ? ☑ YES ☐ NO
Was it detected by Inspector/ Buoff Opr ? ☑ YES ☐ NO

In only one shift ? ☐ YES ☑ NO
In only one line ? ☐ YES ☑ NO
The problem follows the vehicle ? ☑ YES ☐ NO
The problem follows the part ? ☑ YES ☐ NO
Is it a repeat problem ? ☑ YES ☐ NO

Product Failure

Others

Is there a SOS/ SOP / CP/monitoring plan ? ☑ YES ☐ NO
Is it implemented ? ☑ YES ☐ NO
Does it conform ? ☑ YES ☐ N
Is the packaging in good condition ? ☑ YES
Last prevetive measures taken on -
Is there any alert mechanism for failure ? YES ☑ NO
Is the specified product used ? ☑ YES ☐ NO
Did it evolve / Change ? ☑ YES ☐ NO
Is the storage SOP respected ? ☑ YES ☐ NO
Is the shelf life / expiry date respected ? ☐ YES ☑ NO
NA

Material / Consumable

Note : 1. If both fields are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusion (Attach Sketch/ Photograph if necessary) -

1. Defect floows with mirror patch screw.

Is Level 1 analysis required ? ☑ YES ☐ NO

1.Containment Action	**Resp**	**Date**	**Status**
1. New screw bit provided to stage associate once worn out.	Preocess	29.08.2019	Closed

2. Permanent Action -	**Resp**	**Date**	**Status**
1. PART CORRECTION to be done	PVT	30.11.2019	Open

Sustenance Plan for PCA (Audit, Pokayoke)-

Choice : ☐ WSS ☑ QC Story

Analysed by Name/s - Mr. Santosh Jadhav **Date -** 01.08.2019

Inference from L0,
Problem follows part & vehicle
Problem is a repeat problem &
hence detailed analysis required

NO QUICK WIN.

4M analysis Good and Not so good observations	
MAN	Operator Dext. Level L4
Method	Operator aware of key point and check present at buyoff
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Product failure	**Problem follows part**
Material	Handling and shelf life of part adhered

Step 3. Understanding Current Situation

Observation of process:-

TRIM 1 — **TRIMMING + BIW**

TRIM 3 — **DOOR STAGE**
ORVM & Mirror patch assembly on BIW

SUPPLIER
Mirror patch screw

DEFECT GENERATION

DEFECT DETECTION & REWORK

TRIM 3 — **DOOR STAGE**
Mirror patch screw tightening

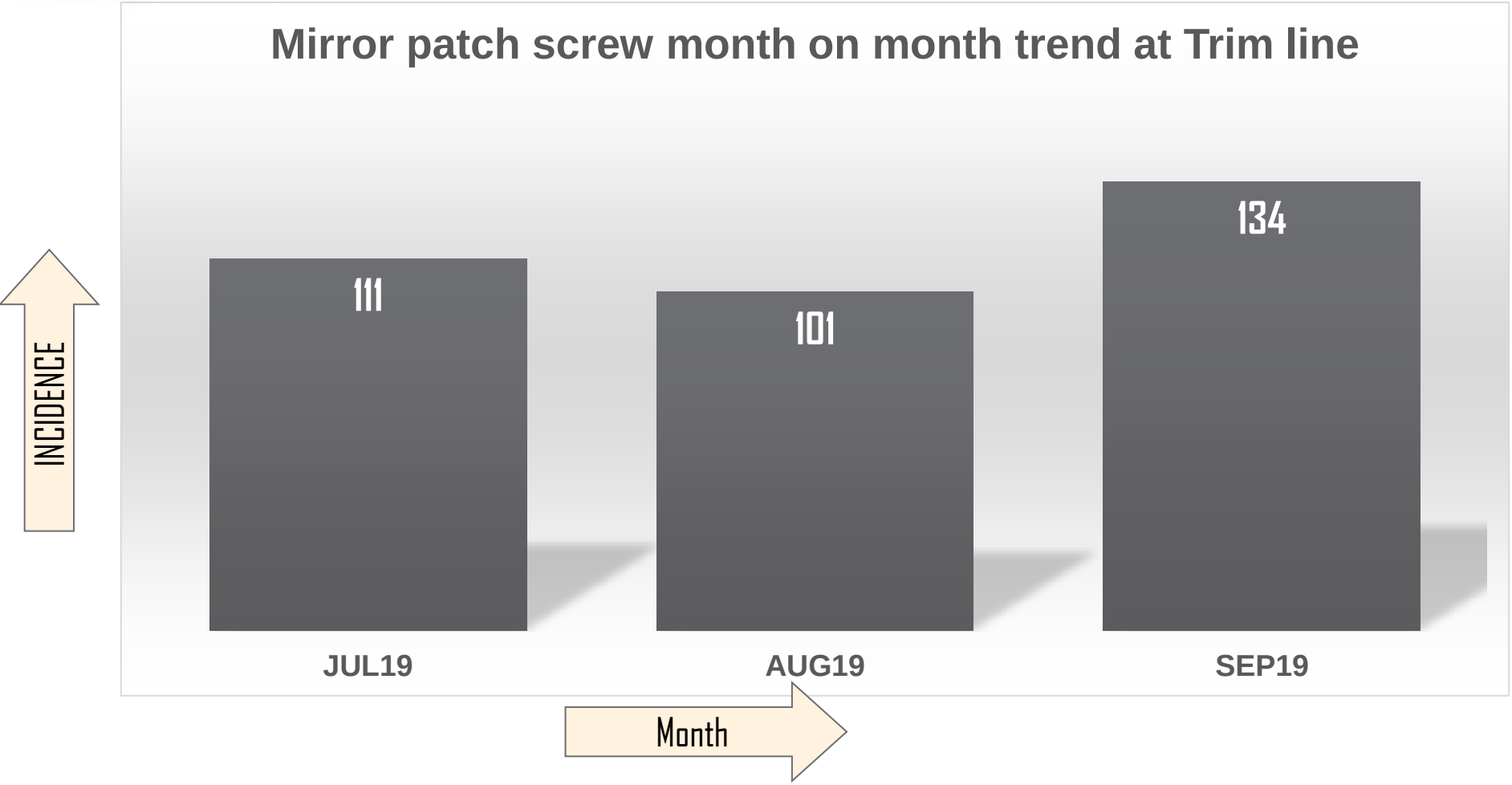
TRIM 3
Checking of screw condition on stage for torque delivery

TRIM 3 Buy off
Defective vehicle rework. OK stamp on Card

FAI
OK to FAI and to CAI for final tag

3. Understand the current situation

Weekly defect trend



INFERANCE:- **Defect incidences are consistent and increased in Sep 19.**

Step 3. Understanding Current Situation

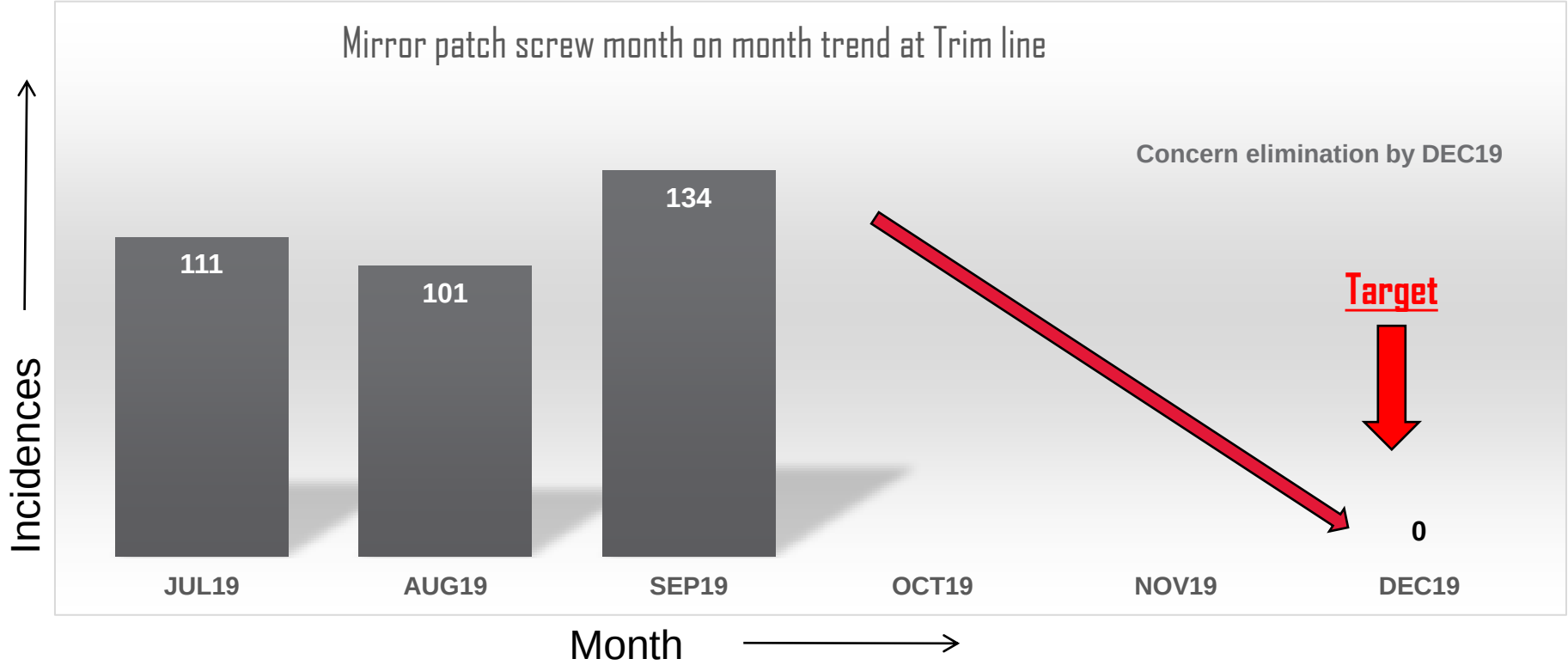
Good & Not so good observations

- Issue observed only in Scorpio Model.
- Issue reported in both the shifts
- Repair process is tricky .
- Issue reported at stage.
- Not all the vehicles having this issue.
- Screw need to damage while removal from vehicle.
- Part getting rejected while rework of defect.
- Not sure about 100% ok condition after rework

4. Select targets

Identify the target. A target is a number indicating the level of improvement to be achieved.

Present Level:	115 Average monthly incidences
Target :	**Concern elimination by DEC2019**

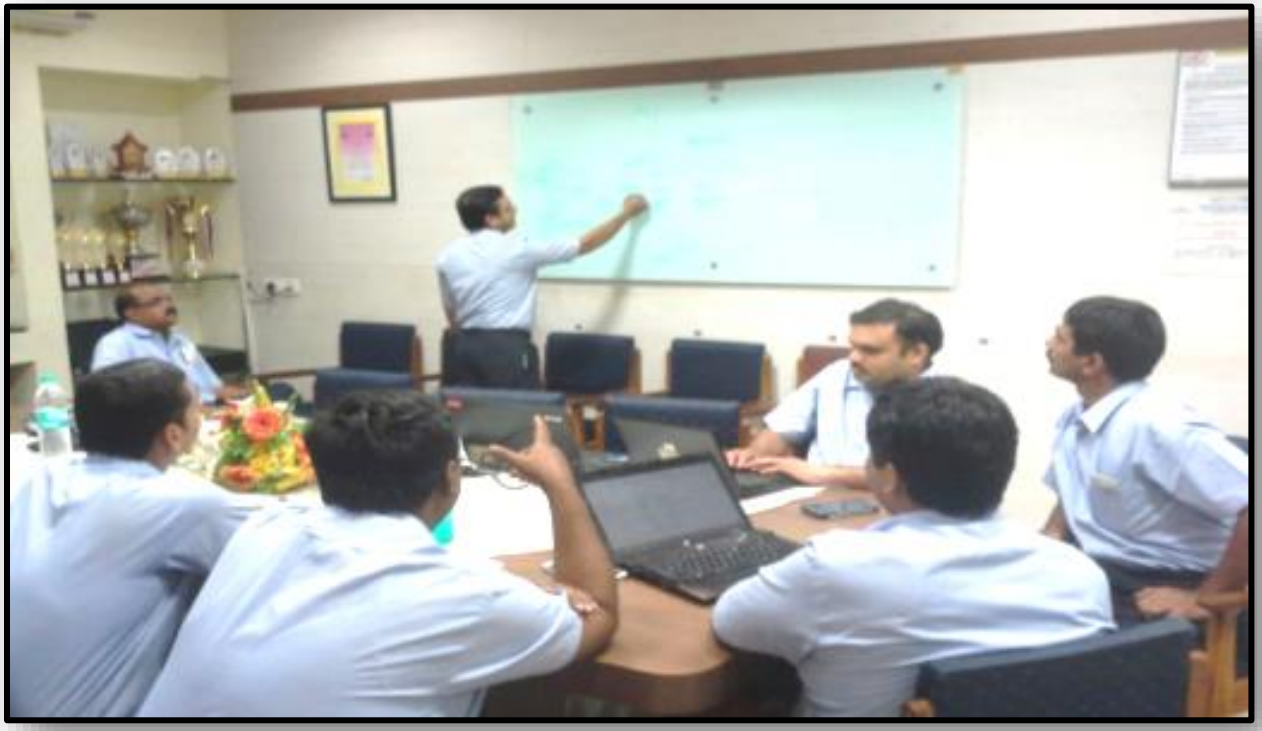


INFERANCE:- Target taken for mirror patch screw slip concern to eliminate concern by DEC 2019.

Step 5. Analysis

List of Probable causes identification in Brain storming session

The possible causes who are having the Total score above 35 or more are considered as the probable causes. In Total of 12 Possible causes 5 Probable causes are identified.



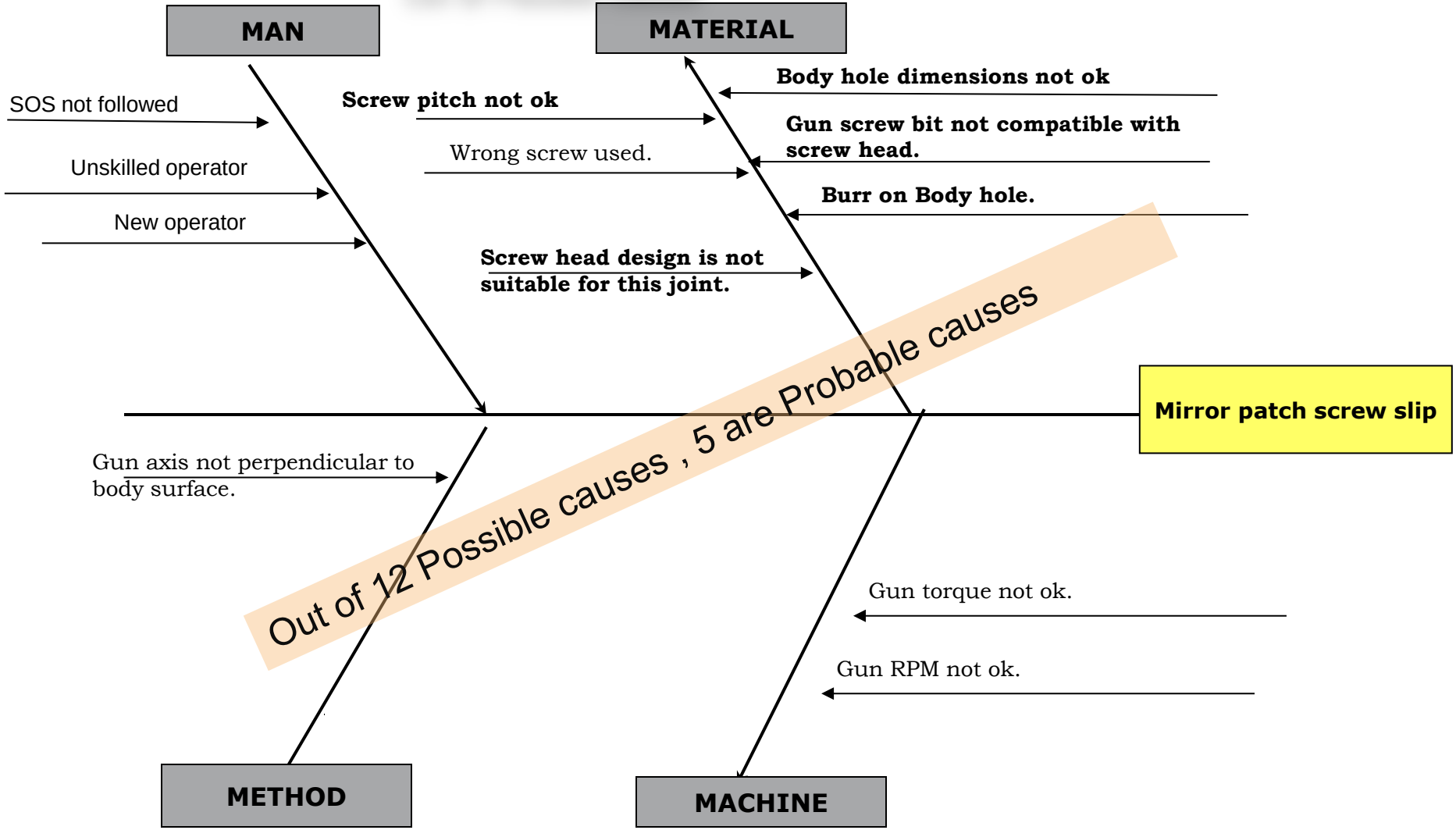
5. Analysis

List of 12 Possible causes after brainstorming among team members

Sr no	Possible cause	Sr no	Possible cause
1	New operator deployed at stage.	7	Gun RPM not ok.
2	SOS Not Followed .	8	Gun screw bit not compatible with screw head.
3	Unskilled operator for VIN plate assembly	9	Wrong screw used.
4	Body hole dimensions not ok	10	Burr on Body hole.
5	Screw pitch not ok	11	Gun axis not perpendicular to VIN plate surface.
6	Gun torque not ok.	12	Screw head design is not suitable for this joint.

Step 5. Analysis

List Of Possible Causes



Step 5. Analysis

List of Probable causes identification in Brain storming session

Sr. no	Possible Causes	Santosh	Satish	Ganesh	Hemant	Sachin	Gobee	Vivek	Total
1	New operator deployed at stage.	1	1	3	3	3	3	3	17
2	SOS Not Followed .	1	1	1	1	1	1	1	7
3	Unskilled operator for VIN plate assembly	3	3	3	3	1	3	3	19
4	Body hole dimensions not ok	9	9	9	3	3	3	3	39
5	Screw pitch not ok	9	9	9	9	1	1	1	39
6	Gun torque not ok.	1	3	1	3	3	1	3	15
7	Gun RPM not ok.	1	1	3	9	1	3	3	21
8	Gun screw bit not compatible with screw head.	9	9	3	3	9	9	3	45
9	Wrong screw used.	1	1	3	3	9	3	3	23
10	Burr on Body hole.	9	3	1	3	9	9	9	43
11	Gun axis not perpendicular to body surface.	1	1	1	3	1	1	1	9
12	Screw head design is not suitable for this joint.	9	9	9	3	9	1	9	49

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting probable cause is 30.

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

Sr.No	Not considered Causes	Reasons
1	New operator deployed at stage.	All trained operator deployed on stage
2	SOS Not Followed .	SOS Followed
3	Unskilled operator for VIN plate assembly	All trained operator deployed on stage
6	Gun torque not ok.	Gun torque checked found ok
7	Gun RPM not ok.	Gun RPM checked found ok
9	Wrong screw used.	Visually checked confirmed OK.
11	Gun axis not perpendicular to VIN plate surface.	Visually checked confirmed OK.

5.1 Identify Probable cause

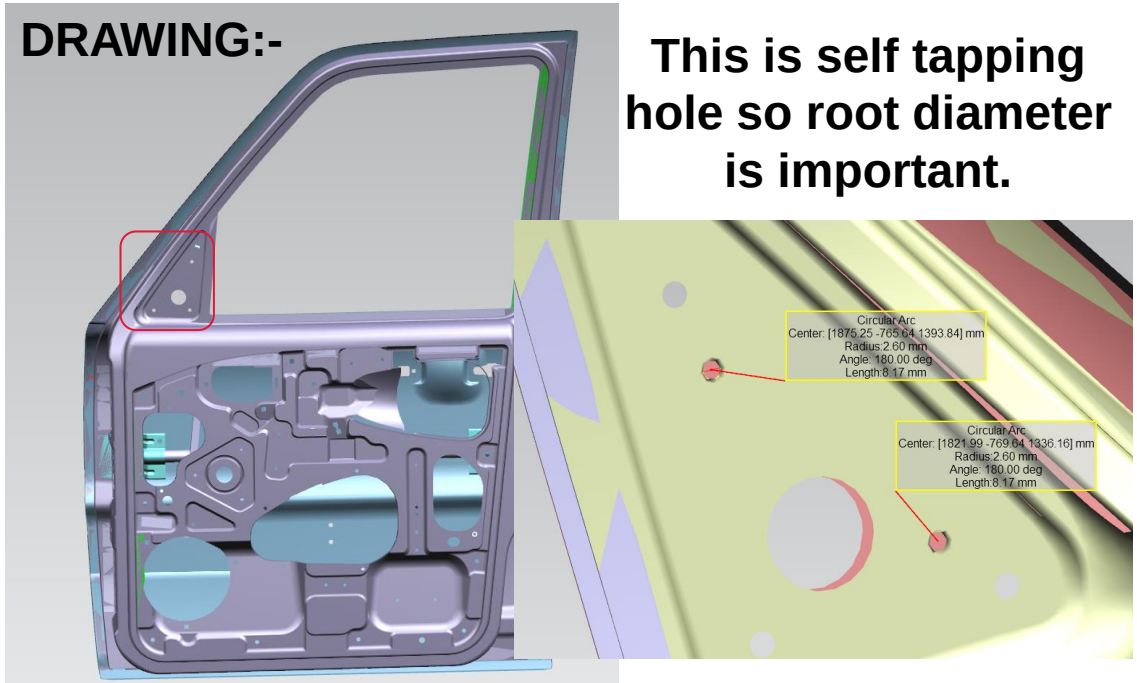
No	List of Probable Causes
4	Body hole dimensions not ok
5	Screw pitch not ok
8	Gun screw bit not compatible with screw head.
10	Burr on Body hole.
12	Screw head design is not suitable for this joint.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Body hole dimensions not ok	L1 analysis report	Body hole is punching operation at press shop-Negligible chances for getting wrong punch size. Checked 10 vehicles for punching hole diameter found OK	**INVALID**

DRAWING:-



This is self tapping hole so root diameter is important.

Specification:-

Body punch hole diameter specification
Ø 5.2 +0.3/-0.0 mm

Observed dimension :-

On defective vehicle-	On BIW level:-
5.2-OK	5.3-OK
5.2-OK	5.2-OK
5.3-OK	5.2-OK
5.2-OK	5.2-OK
5.2-OK	5.2-OK
5.3-OK	5.3-OK
5.3-OK	5.2-OK
5.2-OK	5.3-OK
5.3-OK	5.2-OK
5.3-OK	5.2 - OK
5.2 - OK	

Conclusion:- BIW punch hole dimension not contributing into defect.

TEST OF HYPOTHESIS

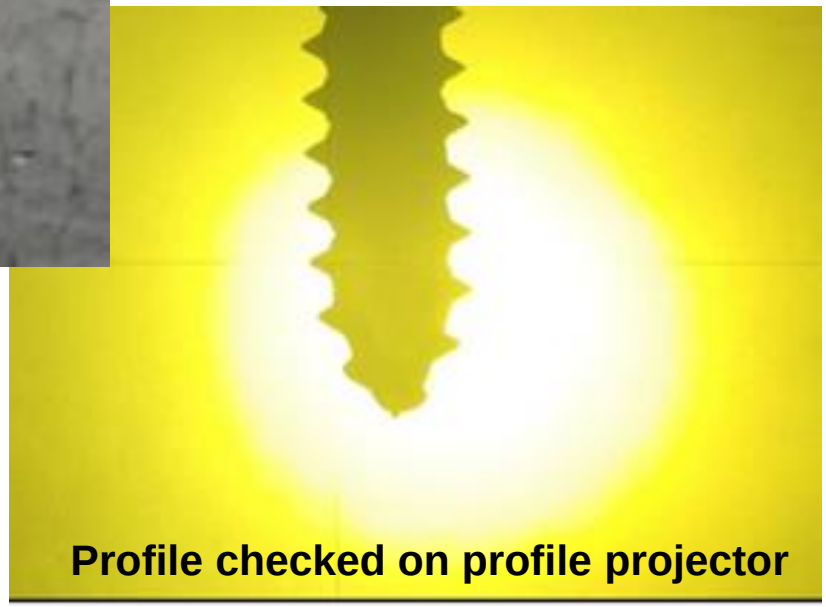
5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
5	Screw pitch not ok	L1 analysis report	Screw pitch is measured on profile projector – found ok.	**INVALID**

Observation :-

1. Both ok and not ok vehicle screw checked on profile projector for major , minor diameter.
2. Found ok.

Conclusion:- Screw pitch and profile is OK



EN14-12047N	A	.	.	SF0216025-NEW RELEASE
NOM SIZE CURRENT				ST4.8
NOM.SIZE FORMER				10
PITCH 'p' (mm)				1.60
UNTHREADED LENGTH MAX 'b1' (mm)				1.60
MAJOR THREADING DIAMETER MAX 'd' (mm)				4.80
MAJOR THREADING DIAMETER MIN 'd' (mm)				4.62
THREAD DIRECTION				RIGHT HAND
SCREW LENGTH 'l' (mm)				25.00
END TYPE				CONE
END SIZE 'y' (mm)				4.30
CROSS RECESS TYPE				H
CROSS RECESS NUMBER				1
CROSS RECESS WING LENGTH 'm' (mm)				4.90
RADIUS MIN 'r' (mm)				0.60
SPHERICAL RADIUS 'R' (mm)				8.00
HEAD DIAMETER MAX 'D' (mm)				11.80
HEAD DIAMETER MIN 'D' (mm)				11.20
HEAD HEIGHT MAX 'k' (mm)				3.25
HEAD HEIGHT MIN 'k' (mm)				2.95
MATERIAL				STEEL
SURFACE FINISH				Fe/Zn 7.5 Tri Block(96wr/240rr)

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
8	Gun screw bit not compatible with screw head.	Standard comparison	Standard studied and as per matrix P2 screw bit to be used – currently we are using same P2 type bit.	**IN VALID**

	SF0216031
NOM SIZE CURRENT	ST4.2
NOM.SIZE FORMER	8
PITCH 'p' (mm)	1.4
MAJOR THREADING DIAMETER MAX 'd' (mm)	4.22
MAJOR THREADING DIAMETER MIN 'd' (mm)	4.04
THREAD DIRECTION	RIGHT HAND
SCREW LENGTH 'l' (mm)	25
END TYPE	CONE
END SIZE 'y' (mm)	3.7
CROSS RECESS TYPE	H
CROSS RECESS NUMBER	2
CROSS RECESS WING LENGTH 'm' (mm)	4.4
RADIUS MIN 'r' (mm)	0.2
SPHERICAL RADIUS 'R' (mm)	6.5
HEAD DIAMETER MAX 'D' (mm)	8
HEAD DIAMETER MIN 'D' (mm)	7.64
HEAD HEIGHT MAX 'k' (mm)	3.1
HEAD HEIGHT MIN 'k' (mm)	2.8
MATERIAL	STEEL
SURFACE FINISH	Fe/Zn 8 Tri Black + TC (96 WR/240 RR)
WEIGHT (Kg)	
ALTERNATE PART NUMBERS	
REMARKS	UNSPECIFIED AS PER 7049

Recommended BIT Size for Size 8

Phillips drive tool and fastener sizes[3][9]

Driver size	Wood screw size	Machine screw size
#0	#0–#1	#0 and #1
#1	#2–#4	#2, #3, #4
#2	#5–#9	#5–#10
#3	#10–#16	#12, $\frac{1}{4}$ in or 6.4 mm, $\frac{5}{16}$ in or 7.9 mm if round-head
#4	#18–#24	$\frac{3}{8}$ in or 9.5 mm, $\frac{9}{16}$ in or 14 mm, plus $\frac{5}{16}$ in or 7.9 mm if flat-head
#5		$\frac{5}{8}$ in or 16 mm, $\frac{3}{4}$ in or 19 mm

Observations:-

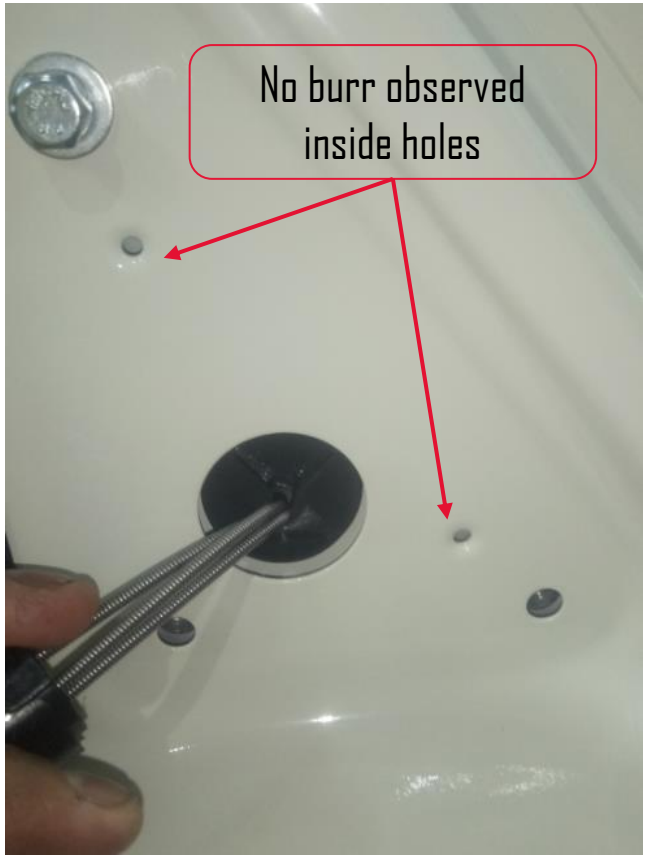
1. P2 screw bit is used on mirror patch screw assembly

Conclusion:- Gun screw bit selection is correct for current screw head.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
10	Burr on Body hole.	L1 analysis report	20 Nos vehicle visually checked for presence of burr on screw mounting area.	**IN VALID**



Observations:-

1. Total 20 nos of BIW checked for screw mounting area burr presence..
2. No defective vehicle found in TCF.
3. Out of this 20 nos 3 vehicle found not ok for Screw digging issue.

Conclusion:- All BIW found ok for burr.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
12	Screw head design is not suitable for this joint.	Design and standard study	Cross recess wing length is required 4.6 which is 4.2 in current design. This recess decided the depth of head.	**VALID**

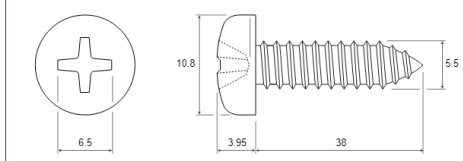
Table according to ISO 7049:

ST 4.2

dimensions in mm

Thread size			ST2,2	ST2,9	ST3,5	(ST3,9)	ST4,2	ST4,8	ST5,5	ST6,3
P			0,8	1,1	1,3	1,3	1,4	1,6	1,8	1,8
a		max.	0,8	1,1	1,3	1,3	1,4	1,6	1,8	1,8
dk		nominal=max.	4,2	5,6	6,9	7,5	8,2	9,5	10,8	12,5
		min.	3,9	5,3	6,54	7,14	7,84	9,14	10,37	12,07
k		nominal=max.	1,8	2,2	2,6	2,8	3,05	3,55	3,95	4,55
		min.	1,55	1,95	2,35	2,55	2,75	3,25	3,65	4,25
r		max.	0,3	0,4	0,5	0,5	0,6	0,7	0,8	0,9
rf		approx.	3,4	4,4	5,4	5,8	6,2	7,2	8,2	9,5
Cross recess		No.	1	1	2	2	2	2	3	3
Type H	m	approx.	2,6	3	4,2	4,4	4,6	5	6,5	7,1
	Penetration deptd	min.	0,86	1,35	1,4	1,63	1,8	2,26	2,49	3
		max.	1,32	1,8	2,03	2,26	2,46	2,87	3,15	3,66
Type Z	m	approx.	2,4	2,9	3,9	4,1	4,3	4,7	6,2	6,7
	Penetration deptd	min.	1,1	1,58	1,47	1,7	1,88	2,28	2,51	3,02
		max.	1,35	1,83	1,93	2,16	2,34	2,74	2,97	3,48
y	max.	Type C	2	2,6	3,2	3,5	3,7	4,3	5	6
		Type F	1,6	2,1	2,5	2,7	2,8	3,2	3,6	3,6

Bit Size as per Recess number 2: PH2



Cross recess wing length.

- **As per ISO 7049**
- Cross Recess Wing Number: 2
- Cross Recess Wing Length: 4.6 mm

Conclusion :- Design is not adequate to match screw head and P2 beat size – this is valid cause

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Conclusion
16	Screw head design is not suitable for this joint.	Design and standard study	**Hypothesis valid**

1 Valid Cause

WHY - WHY Analysis

5.3 Identify the root causes:

No	Valid Probable Cause	WHY	WHY	WHY	ACTION / COUNTERMEASURE
16	Screw head design is not suitable for this joint.	Less screw head depth and recess width	Screw head wing recess not compatible to P2 bit		Screw head to be corrected to make compatible with P2 screw bit.

6. Implement the corrective measures

Possible Remedies Identified

One possible solutions identified for Mirror patch screw slip concern.

1. Screw head to be corrected to make compatible with P2 screw bit.

Legends

High ↑ Low ↓

S N	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	Screw head design is not suitable for this joint.	Screw head to be corrected to make compatible with P2 screw bit.	High ↓	High ↓	No ↓	High ↓	**Accept**

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
1	Screw head design is not suitable for this joint.	Screw head to be corrected to make compatible with P2 screw bit.	PTR started - 10.010.2019 (Implementation of new screw done immediately after PTR)	Vivek Kumar	CLOSED

OBSERVATIONS AND ACTIONS :-

1. Alternate screw identified from bolero shop which is having better screw head depth.
2. Same issue taken up with PVT for ECN change.
3. 1000 parts PTR taken- found ok

Proposed Screw
Part No.- SF0216001
Vender- SHAH Auto

Existing Screw
Part No.- 0105EG0530N
Supplier- PFS

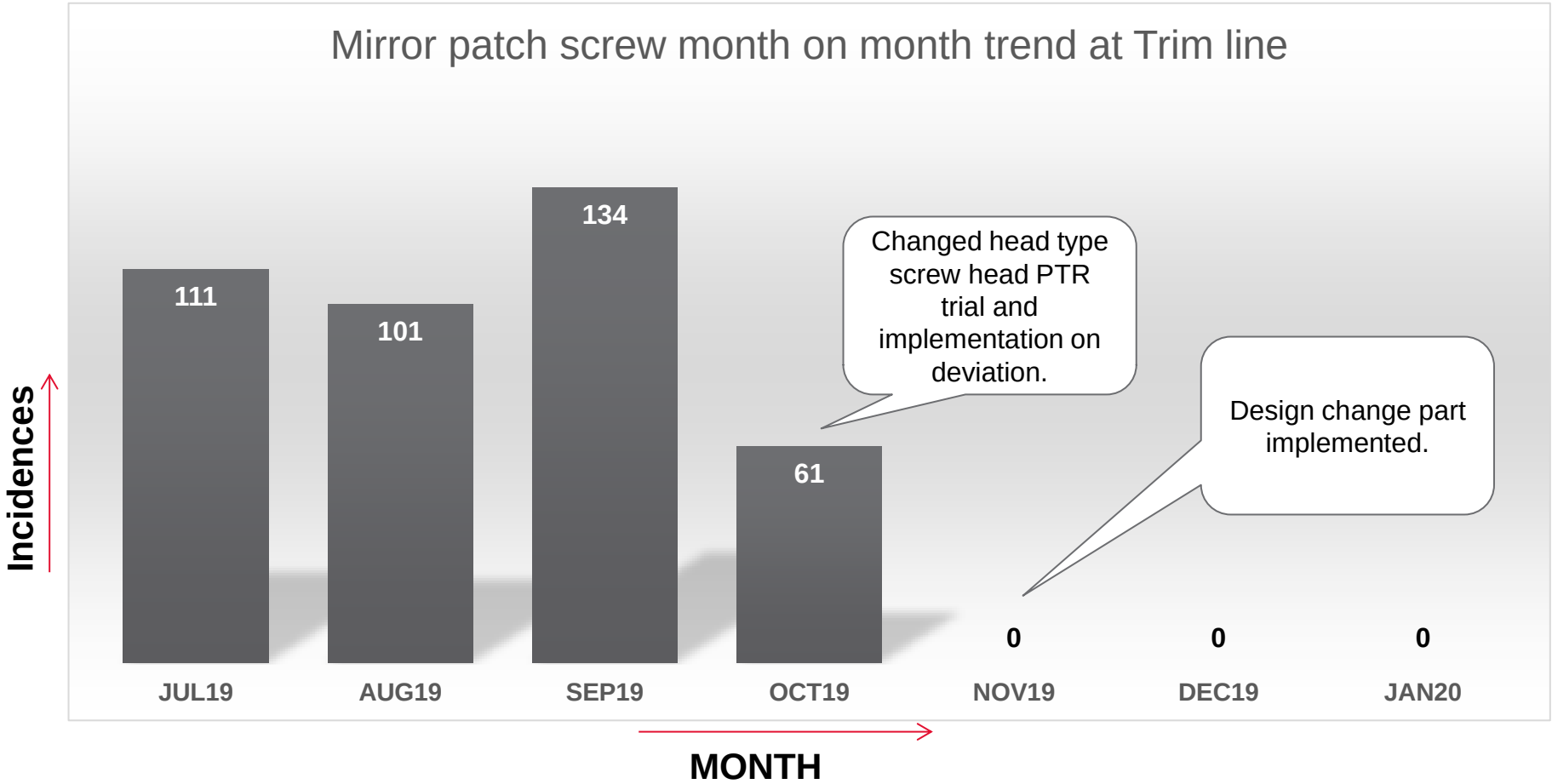
7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Actual date	Responsibility	Status
1	Line DRF for Detection One point lesson for associate	20.07.2019	Ganesh Bendale	CLOSED
2	EN for screw head change.	27.12.2019	Vivek Kumar	CLOSED
3	ADD DELETE mail from R&D to Process.	27.12.2019	Milind Lahire	CLOSED

7. Check & confirm the results & effects

7.1 Verify Results:



INFERENCE:- Mirror patch screw slip issue is eliminated after taking design corrective action.

Check & confirm the results & effects (Cont..)

7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) Elimination of mirror patch screw slip issue.
2) 132414 Rs Cost Saved/ Annum
3) FRC improved by 1% from 0 %
4) Repeat and chronic concern for bit rejection

In tangible Benefits :-

1) Improved Customer Satisfaction.
2) Major probable field issue protected to pass on.
3) Operator fatigue reduced , moral improved

Wed 1/15/2020 7:57 AM

DHANOKAR RAJESH

Re:COPQ cost for MYB

To JADHAV SANTOSH--NASHIK; DHANOKAR RAJESH

okay approved

Sent from my Redmi Note 4X

On JADHAV SANTOSH--NASHIK <JADHAV.SANTOSH8@mahindra.com>, 14-Jan-2020 7:44 PM wrote:

Dear sir,

Please approve COPQ cost for MYB project.

2. Justify the choice

Cost of Poor Quality

Tightening gun bit :35,640 Rs/-

Cost of 1 Screw bit = 99 Rs per piece
Bit consumption increased by 30 nos per month
Total cost = 99 *30*12 = 35,640 Rs /-
Rejection projection in one year =35640 Rs / Year

Rework Cost estimation for handbrake bolt half torque-

Description	Units	
Rework time (Approx)	Min	10
Avg. Concerns/Day	Nos	10
Total Required Rework time per day	Min	100
Effective working time	Min	465
Operator required /day	nos	0.22
Approx Operator cost/ day	Rs	1500
Operator cost /concern	Rs	322.58
Avg no. of working days	days	25
Operator cost /month	Rs	8064.52
Operator Cost/year	**Rs**	**96774.19**
Screw bit cost	Rs / month.	2970
yearly cost Tap cost	Rs / month.	35640
Total Rework cost	Rs / Year	132414

Total Cost/ year = Material cost + Manpower cost = 35640+96114 = 1,32,414/-

Mahindra Quality System (A. S.)

7

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
1	Line DRF for Detection.	20.07.2019	Ganesh Bendale	CLOSED
2	EN for screw change	27.12.2019	Vivek Kumar	CLOSED
3	New screw drawing	27.12.2019	Vivek Kumar	CLOSED
4	FTR , DCR	27.12.2019	Vivek Kumar	CLOSED

8.2 Training:

SN	Training Identified	Actual date	Responsibility	Status
1.	Training with SOS to line associates	10.01.2020	Ganesh Bendale.	Closed

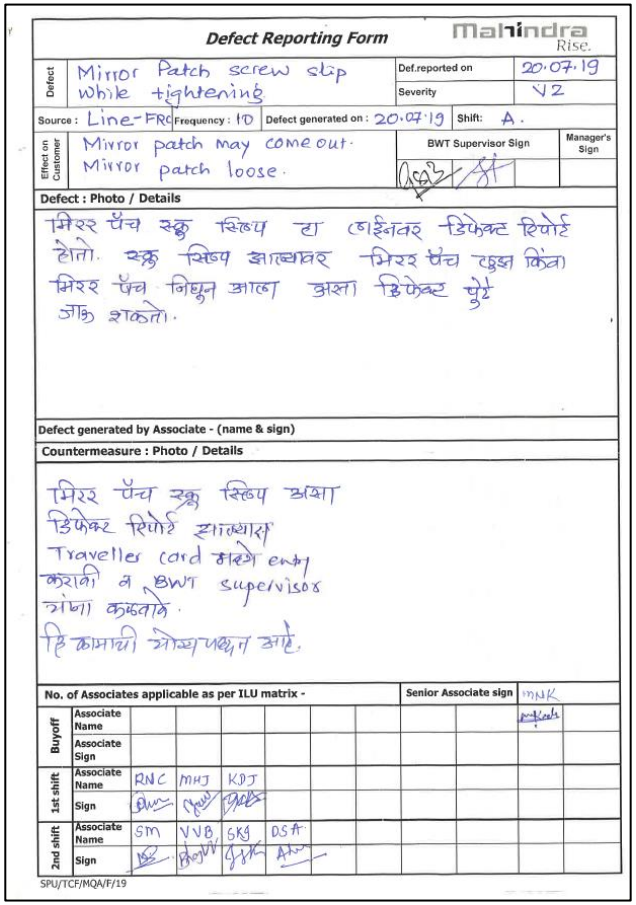
8.3 Additional Audit & check

1	Check point added in Wrap check sheet	17.12.2019	Santosh	Closed

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
1	Line DRF for Detection.	20.07.2019	Ganesh bendale	CLOSED



Defect Reporting Form — Mahindra Rise.

Defect	Mirror Patch screw slip while tightening	Def.reported on	20.07.19
		Severity	V2
Source: Line-FRC	Frequency: 10	Defect generated on: 20.07.19	Shift: A.
Effect on Customer	Mirror patch may come out. Mirror patch loose.	BWT Supervisor Sign	Manager's Sign

Defect : Photo / Details

मिरर पॅच स्क्रु स्लिप हा लाईनवर डिफेक्ट रिपोर्ट होतो. स्क्रु स्लिप झाल्यावर मिरर पॅच खुला किंवा मिरर पॅच निघून आला असा डिफेक्ट पुढे जाऊ शकतो.

Defect generated by Associate - (name & sign)

Countermeasure : Photo / Details

मिरर पॅच स्क्रु स्लिप असा डिफेक्ट रिपोर्ट झाल्यास Traveller card मध्ये entry करावी व BWT supervisor यांना कळवावे.
हि कामाची जोखीमपद्धत आहे.

No. of Associates applicable as per ILU matrix -									Senior Associate sign	MNK
Buyoff	Associate Name									
	Associate Sign									
1st shift	Associate Name	RNC	MHJ	KDJ						
	Sign									
2nd shift	Associate Name	SM	VVB	SKG	DSA					
	Sign									

SPU/TCF/MQA/F/19

Conclusion:-

1. Mirror patch screw head slip is observed after assembly is made aware to line associate.
2. Same is to be reworked on stage.

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
2	EN for new screw head.	27.12.2019	Vivek Kumar	CLOSED

Mahindra & Mahindra Ltd.
Product Development

"Confidential Report"

Page No 1

Mahindra Rise.

Engineering Notice Release: W110-EN-A003-000229
AD Product Development
Mahindra & Mahindra LTD(A.S)

Created Date 02-Jan-2020
Initiated Date 13-Jan-2020

Engineering Notice No : W110-EN-A003-000229

Released Date : 14-Jan-2020 15:34

Reference Change Req No : DCR045475

Location : AD-NASIK

Implementation Location : M&M-Nasik

Engineering Notice Description : MIRROR PATCH SCREW CHANGE FROM #0105EG0530N TO #SF0216001

Reason : TO ADDRESS ISSUE OF SCREW HEAD DIGGING & TOOL BIT REJECTION AT SCORPIO TCF

Quantity : -(Regular)

Reason Code : Quality Improvement

Priority : Urgent

Note : TO BE IMPLEMENTED FOR PRODUCTION

Affected Models : SCORPIO W110 ALL, SCORPIO SC/DC ALL

Is EN Re Released? :

Parent EN NO :

Life Cycle State : Released

Owner Name : MILIND RAMCHANDRA LAHIRE (23075413)

Engineer : MILIND RAMCHANDRA LAHIRE (23075413)

Approved By : 211714-PATHAK YOGESH

Authorized By : HEAD-PVT-AD_202475-RAUT AVESH

Legends

Add New : A Part/Assy that is getting released through ECN for the very first time

Modified : A Part/Assy that is already released through an ECN and has undergone some changes

Added with Modification : It is very similar to "Modified" status except that it is being added newly into a different Parent

Existing : A Part/Assy that is already being used is carried forward as it is to some other parent

Deleted : A Part/Assy that is being used is removed from a Parent

Reinstated : Reintroduction of a Part/Assy that was used Previously in M&M but was discontinued

Added/Deleted Qty : It's delta of existing released EN for the same parent part number

ICA :
O - New part number can be used in place of old part no. & vice versa;
1 - New part number can be used in place of old part no. but not vice versa;
2 - Old part number can be used in place of new part no. but not vice versa;
X - No ICA between new and old part numbers;
Y - Interchangeability as set of parts (to use new part no. we have to use more than one new part no);
NA - Not Applicable;

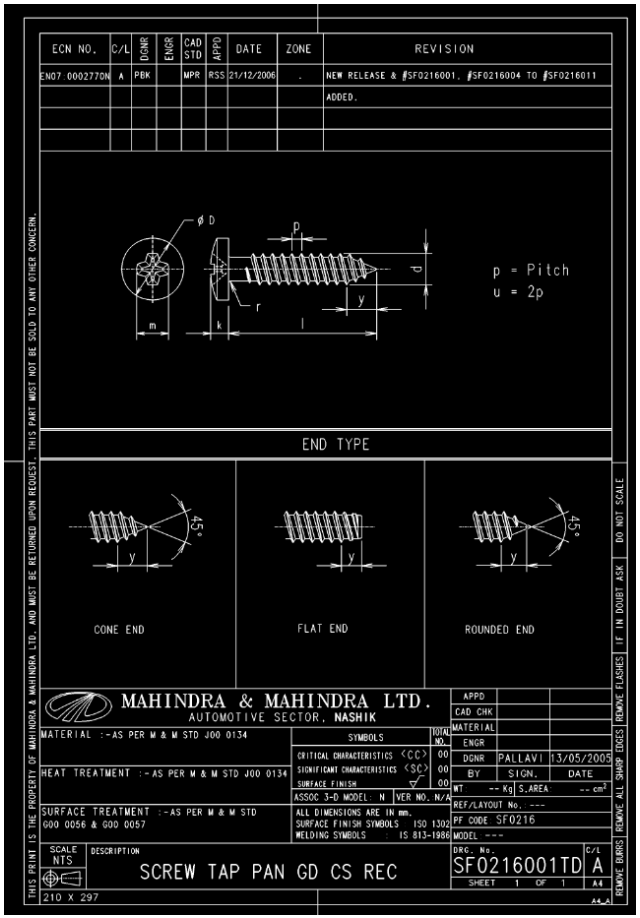
W110-EN-A003-000229

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
3	New screw drawing	27.12.2019	Vivek Kumar	CLOSED

NOM SIZE CURRENT	ST4.2	
NOM.SIZE FORMER	8	
PITCH 'p'	1.40000	mm
MAJOR THREADING DIAMETER MAX 'd'	4.20000	mm
MAJOR THREADING DIAMETER MIN 'd'	4.04000	mm
THREAD DIRECTION	RIGHT HAND	
SCREW LENGTH 'l'	9.50000	mm
END TYPE	CONE	
END SIZE 'y'	3.70000	mm
CROSS RECESS TYPE	H	
CROSS RECESS NUMBER	1	
CROSS RECESS WING LENGTH 'm'	4.60000	mm
RADIUS MIN 'r'	0.20000	mm
SPHERICAL RADIUS 'R'	0.00000	mm
HEAD DIAMETER MAX 'D'	12.00000	mm
HEAD DIAMETER MIN 'D'	11.50000	mm
HEAD HEIGHT MAX 'k'	3.05000	mm
HEAD HEIGHT MIN 'k'	3.00000	mm
MATERIAL	STEEL	
SURFACE FINISH	Fe/Zn 8 Tri Silver Blue + TC (120 WR/240 RR)	
WEIGHT (kg)		kg
ALTERNATE PART NUMBERS	---	



8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
4	FTR , DCR , Add delete	27.12.2019	Vivek Kumar	CLOSED

Mahindra & Mahindra Ltd.

Mahindra Rise. **DESIGN CHANGE REQUEST**

DCR Number	DCR045475
Source Of Change Request	202475 - RAUT AVESH
Product Owner	202475 - RAUT AVESH
Sub Product Owner	211714 - PATHAK YOGESH
Location	AD-NASIK
Reason Code	Quality Improvement
Change Description	MIRROR PATCH SCREW CHANGE FROM #0105EG0530N TO #SF0216001
Why this change ?	TO ADDRESS ISSUE OF SCREW HEAD DIGGING & TOOL BIT REJECTION AT SCORPIO TCF
Proposed Change	MIRROR PATCH SCREW CHANGE FROM #0105EG0530N TO #SF0216001
Priority	Normal
CAE Required	No
CAE	
DFMEA/QFD Required	Yes
Designer	23075413 - LAHIRE MILIND
Lifecycle Status	Approved

Estimated Cost Impact

Type	Saving	Increase
Material		
Conversion		
ManPower		

Affected Model/Major Agg List

SCORPIO ALL

Approval Statge	Approver	Role	Status	Approval Date	Remarks
Initiation	23075413 - LAHIRE MILIND	Initiator	Initiated	05-DEC-2019	
Preliminary Approval by SPO	211714 - PATHAK YOGESH	SPO	Approved	05-DEC-2019	
Work Assigned to DFMEA/QFD	23075413 - LAHIRE MILIND	DFMEA	Completed	05-DEC-2019	DFMEA ATTACHED
Work Assigned to DFMEA/QFD	23075413 - LAHIRE MILIND	DFMEA	In Progress		
DFMEA/QFD Review by SPO	211714 - PATHAK YOGESH	SPO	Approved	05-DEC-2019	
ECN Release	23075413 - LAHIRE MILIND	DESIGNER	In Progress		

Dec 8, 2019, 3:23 PM

FTR/STCF/F20/606.

Mahindra Rise — Fitment Trial Report

Format No.: MQS/PP/P-Mfg/F/02
Rev. Date: 05/09/2019
Rev. No.: 06

Project Name/ Code:

Type of Fitment Trials: ☐ Photo FTR (FTR 1) ☐ Supplier FTR (FTR 2) ☐ Inhouse FTR (FTR 3)

Trigger for Trials: ☐ New ☐ CAN ☐ Prior ☐ Others (specify)

Trial Report Serial No:- | Date:-

From (Requesting Dept): STCF | To (Trial Executing Dept.): PUY

Part Number :- 0105EG0530N | Part Description :- Mirror patch screw - TRUSS SCREW

Vendor Name :- | Feedback expected from any other dept :- –

Model :- SCORPIO-ALL | Is change required in VAHAN database ? Yes / No NO

DCR/PPRF/EN No :- | Trial Quantity :- 50

Change Details

i) Description Of Change :- Alternate mirror patch screw to be used. CURRENT-0105EG0530N proposed-SF0216001

ii) Reason for Change :- Existing screw head material digging and bit rejection in TCF.

iii) Acceptance Criteria :- No head slip and bit damage while torquing.

FTR Raised By :- Jothav Santosh. Sign:- Date:- 21.11.2019.

Comments to be obtained from NOVA-C & RFI ☐ Yes ☐ NO

CONSENT BY QA (Manager- Vehicle) for the trial : Sign :- Date :- 21/11/19

Vehicle Details: 1DO NOP

Vehicle Serial Nos :- | Aggregate Serial Nos :-

Comments On Fitment

Note: Please ensure that part inspection report(s) (with actual dimns) is attached.

1. Buyer Development Engineers — Accepted ☐ Yes ☐ NO
Comments:
Name :- | Sign :- | Date :-

2. Supplier Quality Engineering (SQE) — Accepted ☐ Yes ☐ NO
Comments:
Name :- | Sign :- | Date :-

3. Supply Quality Assurance (SQA) : — Accepted ☐ Yes ☐ NO
Comments:
Name :- | Sign :- | Date :-

4. Manufacturing Shop / Process Engineer — Accepted ☐ Yes ☐ NO
Comments: Fitment issue No issue.
Any Safety/Environmental related issue observed while fitment of Part ☐ Yes ☐ NO
Name :- A.B. Konwade | Sign :- | Date :- 21.11.19.

5. Manufacturing Quality Assurance (Respective Area) — Accepted ☐ Yes ☐ NO
Comments: Head and bit compatability - OK. No Head slip observed. to be implement.
Comments Required: ☐ ~~GNOVA-C~~ ☐ RFI Incharge
Name :- Jothav Santosh | Sign :- | Date :-

GNOVA-C/ RFI Incharge — Accepted ☐ Yes ☐ NO
Comments:
Sign :- | Date :-

RFI Incharge — Accepted ☐ Yes ☐ NO
Comments:
Sign :- | Date :-

Is the change carried out in samples is interchangeable with old vehicles? ☐ Yes ☐ NO

Whether non Interchangeable parts removed from trial vehicles? (Mandatory to remove if it is non Interchangeable part) ☐ Yes ☐ NO

Head PV/TP/DT/PET/Vehicle Engineering — Sign : Date :

Head Product Quality — Sign : Date :

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
4	FTR , DCR , Add delete	27.12.2019	Vivek Kumar	CLOSED

W110-EN-A003-000229

ADD / DEL LIST

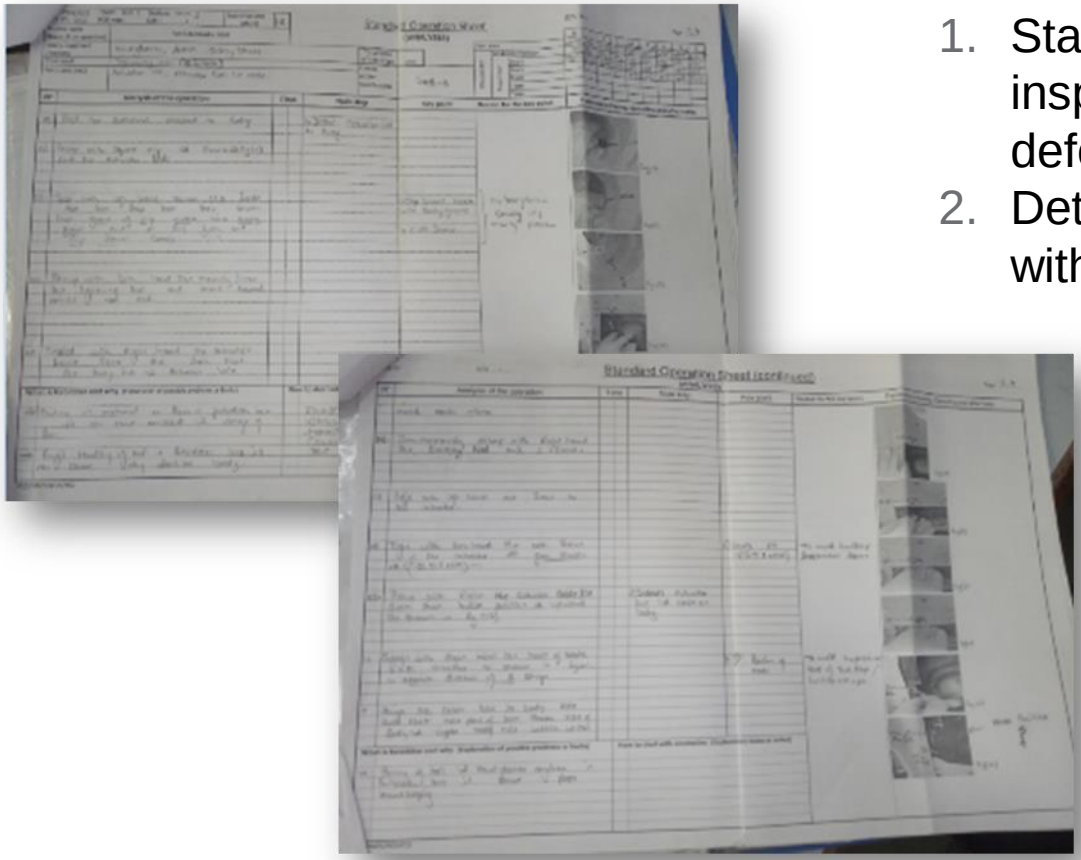
DCR045475	EN NO: W110-EN-A003-000229	
Description:	MIRROR PATCH SCREW CHANGE FROM #0105EG0530N TO #SF0216001 (Already using in BOLERO for mirror patch screw).	

ADD						DELETE				Model / Remarks
Part No.	Rev	Description	Qty.	End Item (Y/N)	Drg (M/S)	Part No.	Rev	Description	Qty	
SF0216001	A	SCREW TPPA CR ST4.2X1.4X9.5 C WH	4	Y	M	0105EG0530N	B	ST TRUSS HD SCREW NO.10X10AB TYP	4	SCORPIO W110 ALL, SCORPIO SC/DC ALL

8. Standardization

8.2 Training:

Action Plan	Target Dt	Resp.	Status
Training with SOS to line associates	10.01.2020	Ganesh Bendale.	Closed



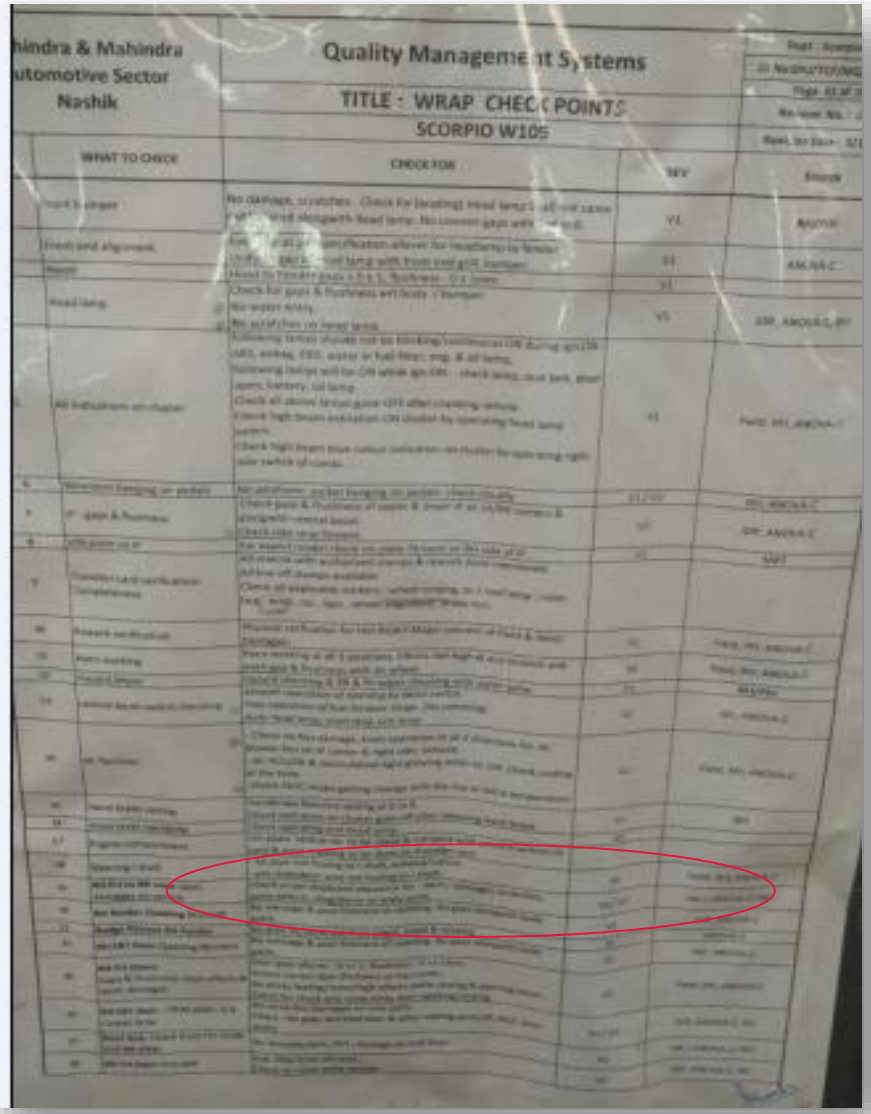
1. Stage associate and buy off inspectors are made aware about defect reported.
2. Detection and prone causes shared with line associate.

8. Standardization

8.3 Additional Audit & check

Wrap check sheet updated

Wrap check sheet is updated for dynamic check for defect for time being.



9. Summarise & Plan future actions

9.1 Conclusion:

After taking corrective action in mirror patch screw head issue is eliminated from TCF.

9.2 Horizontal Deployment:

Scuff plate screw head slip – is same kind of defect is getting reported in Scorpio model in TCF

9.3 Lessons Learnt:

Things Gone Wrong :

Supplier recommendation and mating part geometry not studied while design, Other model is also not benchmarked for same part fitment.

Things Gone Right :
Design and fastener standard study always plays important role in problem solving.

Supplier recommendation taken into consideration while analysis,Gemba observations plays important role in problem solving.

Thank you

Visit us at www.mahindra.com

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